Department of Chemistry
COLORADO SCHOOL OF MINES
Name: **Andrew Yonan**
Aspirin Sample Code: **BLUE**
Section: **LAA**

Experiment 6 (Aspirin Titration)

1.

Trial #	Weight KPH (g)	Volume NaOH added (mL)	Concentration NaOH (M)
1	0.3064	15.92	0.09425
2	0.3062	15.67	0.09569
3	0.3062	15.61	0.09606

2. The NaOH standardization reaction is as follows:

$$\underbrace{C_6H_4(COOK)(COOH)}_{KHP} + NaOH \implies H_2O + C_6H_4(COOK)(COONa)$$

The molar mass of KHP is 204.2 g/mol. We see that 1 mole of KHP reacts with 1 mole of NaOH. Hence, for trial 1,

$$\frac{0.3064}{204.2} = 0.001500 \text{ mol}$$

of KHP must have reacted with the same moles of NaOH. Hence the concentration of NaOH in this trial is

$$\frac{0.001500 \text{ mol}}{0.01592 \text{ L}} = \boxed{0.09425 \text{ mol/L}}$$

3. The average NaOH molarity is

$$\bar{x} = \frac{0.09425 + 0.09569 + 0.09606}{3} = \boxed{0.09533 \text{ M}}$$

4. The sample variance of a data set $\{x_1, x_2, \cdots, x_n\}$ is given by the unbiased estimator

$$s^2 = \frac{1}{N-1}\sum_{i=1}^{N}(x_i - \bar{x})^2$$

Where N is the number of trials, x_i is the ith value, and $\bar{x}$ is the average value. Hence, the standard deviation is

$$s = \sqrt{\frac{(0.09425 - 0.09533)^2 + (0.09569 - 0.09533)^2 + (0.09606 - 0.09533)^2}{2}}$$
$$= 0.00096$$

5. The relative standard deviation is

$$100\left|\frac{s}{\bar{x}}\right| = 1.0\%$$

6.

Trial #	Mass Aspirin Powder (g)	Volume NaOH (mL)	Mass Pure Aspirin (g)
1	0.5007	22.96	0.3943
2	0.5005	22.61	0.3883

7. The reaction of Aspirin with NaOH is given by

$$C_9H_8O_4 + NaOH \implies C_9H_7O_4Na + H_2O$$

Hence 1 mole of Aspirin reacts with 1 mole of NaOH, and hence the moles of Aspirin must be equal to the moles of NaOH. Then, for trial 1, using our standardized NaOH concentration, we have

$$(0.02296\ \text{L})(0.09533\ \text{mol/L}) = 0.002189\ \text{mol}$$

of Aspirin, which translates to 0.3943 grams (the molar mass of Aspirin is 180.159 g/mol).

8. We calculate the mass percent of pure Aspirin in the Aspirin powder for both trials:

$$p_{\text{aspirin, trial 1}} = \frac{0.3943}{0.5007} = 78.75\%$$

$$p_{\text{aspirin, trial 2}} = \frac{0.3883}{0.5005} = 77.58\%$$

9. The average mass percent Aspirin in the unknown sample is then

$$p_{\text{aspirin}} = \frac{78.75 + 77.58}{2} = \boxed{78.17\%}$$

10. Three potential sources of error are:

    (a) wind- / air-flow-corrupted 4-place balance reading (mitigated by shutting the door on the 4-place balance)

    (b) improper weight paper handling, causing grains of weighed material to sprinkle off (mitigated by placing the material in the Erlenmeyer flask immediately upon taking the mass reading)

    (c) residual wall-sticking NaOH during titration, causing less NaOH to react than was recorded leaving the buret (mitigated by washing down the sides of the Erlenmeyer flask with DI water during titration)

11. Overflowing the buret in a way that causes some NaOH to slip into the acid means the reaction starts further en route to the endpoint than expected (if the expected titration progress is 0). Hence, it will take less quantity (volume) of NaOH to reach the endpoint than it would have, had the reaction not been head-started by the overflow. Since the concentration of NaOH in the standardization reaction is given by

$$C = \frac{\text{grams of KHP}}{(\text{Molar mass KHP})(\text{Volume NaOH})}$$

Less NaOH means higher C. Hence, the reported concentration will be **higher**.

12. If some of the acid doesn't react with the base because it's still trapped in undissolved solid, then you have less reaction-contributing acid than recorded. Because less acid means less base is required to titrate, you end up needing less base to reach the endpoint than the recorded acid mass would have predicted. Because the calculated acid mass is proportional to the volume of consumed base, the calculated acid mass becomes lower—which also **lowers** the calculated concentration of acid in a fixed volume.

Because the calculated acid mass decreased and we (presumably) use the unchanged total mass of acid solid (including the undissolved acid) as the denominator of our percent fraction, the reported percent mass of acid will be **lower**.